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# Tracheostomy in children: Recommendations for a safer technique

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ABSTRACT

Over the last few decades, tracheostomy has been increasingly performed in children with various complex and chronic conditions. We have seen a dramatic change in indications for tracheostomy in pediatric patients due to better survival of premature infants and those suffering from severe congenital anomalies. There is no consensus about the timing of tracheostomy in pediatric patients. Although percutaneous tracheostomy has become the standard of care for adults, there is not sufficient evidence to start performing it routinely in pediatric patients. The indications, preoperative considerations, and different procedures for tracheostomy in children are reviewed. Surgical tracheostomy is described step by step placing an emphasis on safety measures to minimize complications. There is also a great variability in tracheostomy care protocols in the literature. Post-operative tracheostomy care is discussed for the early and late post-operative periods. There is no general consensus on decannulation protocols, but prevailing expert opinion is presented. There is growing evidence in support for an interdisciplinary approach to pediatric tracheostomized patients. The focus of this paper is the review of the literature regarding safety improvement strategies from the surgical and post-operative point of view.

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## Introduction

Over the last few decades, tracheostomy has been increasingly performed in children with various complex and chronic conditions. We have seen a dramatic change in indications for tracheostomy in pediatric patients due to better survival of premature infants and those suffering from severe congenital anomalies. Previously, the main indication for pediatric tracheostomy was airway obstruction due to infectious diseases[1], but now, most tracheostomies are being performed for prolonged ventilation, acquired or congenital airway stenosis, and craniofacial abnormalities. More than 55% of children are under the age of 1 year at the time of tracheostomy. Decannulation rates for these children are extremely low, ranging from 28 to 51%, and in those children who are decannulated, the average time the tracheostomy is present is 2 years.[2] This shift also means that tracheostomized patients are more complex than ever, and require the care of an interdisciplinary team to address the multiple issues associated with their underlying pathologies. Of 206 children who underwent an elective tracheostomy from 2012 to 2013, 34% were neonates, 54.4% were born prematurely, 97.6% were categorized as American Society of Anesthesiologists class 3 or higher, 98.5% had pulmonary risk factors, and 75.7% required nutritional support.[3] For many of these medically complex children, the timing of the tracheostomy and the preoperative discussion regarding ongoing care is significantly challenging.[4]

## Indications for tracheostomy in children

Albeit tracheostomy indications have broadened in the past few decades, it is indicated basically for the same reasons as before: either it serves as a bypass to an obstruction in the airway, or it is instituted for long-term respiratory support and/or pulmonary toilet.[5]

During the early years, the technique was used mainly for acute airway obstruction secondary to infectious diseases such as diphtheria or epiglottitis. However, less than a third survived, and many parents would not allow their children to have a tracheostomy.[6] This only changed when Chevalier Jackson standardized the procedure, reducing morbidity and mortality significantly.[7] In the years to come, the list of indications for tracheostomy gradually increased, and it gained notoriety with the infamous Copenhagen poliomyelitis epidemic of 1952.Tracheostomy became standard of care for basically all patients with respiratory insufficiency.[8]

In the past few decades, the range of conditions that may warrant a tracheostomy have broadened significantly. Despite a decrease in emergent tracheostomies, the incidence of tracheostomies in children has not fallen; instead, it has risen especially in tertiary institutions. This has largely been attributed to increased survival of patients in pediatric and neonatal intensive

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**Table 1**
Indications for tracheostomy.

| |
|---|
| **Upper airway obstruction** |
| Craniofacial anomalies |
| Craniofacial/airway trauma/burns |
| Obstructive pharyngeal masses |
| Bilateral vocal fold paralysis |
| Laryngeal web/atresia |
| Subglottic stenosis acquired or congenital |
| Acquired tracheal stenosis |
| Tracheobronchomalacia |
| Extrinsic compression (vascular, tumoral) |
| Obstructive sleep apnea |
| **Long-term ventilation/pulmonary toilet** |
| Chronic pulmonary disease |
| Neurological / Neuromuscular impairment congenital or acquired |

care units. As a result, pediatric tracheostomy is now most commonly performed in children with chronic ventilator dependence, congenital or acquired upper airway anomalies and neurological impairment.[9]

Most frequent conditions that may require a tracheostomy are summarized in Table 1.

### *Tracheostomy for upper airway obstruction*

Today, most causes of emergent tracheostomy can be treated by oropharyngeal intubation. The rare cases where emergency tracheostomy is necessary, usually involve laryngeal/tracheal/midface trauma, or upper airway congenital anomalies without prenatal diagnosis. The main indication for tracheostomy in acquired upper airway obstruction has shifted to causes allowing an elective intervention, such as acquired or congenital subglottic stenosis, bilateral vocal cord paralysis, severe tracheomalacia, and upper airway trauma. These tracheostomies are often temporary measures until corrective surgery can be performed, or the condition no longer generates significant airway obstruction.

### *Tracheostomy for long-term ventilation/pulmonary toilet*

Although there is consensus that tracheostomy has to be performed in 1 or 2 weeks of ventilation in adult patients, no established criteria currently exist regarding time to tracheostomy for children, and thus each patient is evaluated individually. Newborns can tolerate longer periods of intubation (>50 days) without experiencing serious laryngeal adverse effects. A study in 2002[10] showed that children aged 1 month and older tolerate prolonged intubation without serious laryngeal complications, and they found no evidence linking the length of intubation to the probability of tracheostomy. Whether this holds true in burn patients remains controversial. An adverse impact of prolonged intubation has been described in 36 severely burned pediatric patients, in whom transition to tracheostomy before day 10 after the burn was associated with less subglottic stenosis (15%) compared with delayed tracheostomy from day 10 onwards (>50%).[11] On the other hand, Holloway et al[12] analyzed 73 subjects with a median of 22 days of ventilation before tracheostomy, and results suggested that a longer duration of ventilation before tracheostomy is associated with increased ICU morbidities and stay. Early tracheostomy was recommended, suggesting that it may have significant benefits without adversely affecting mortality. Thus, after 2 weeks of intubation in a child, one should consider tracheostomy evaluation, provided the child is stable on the ventilator.

## Preoperative considerations

Tracheostomy is being increasingly performed in children with complex chronic conditions. Of a retrospective cohort of 502 children who underwent tracheostomy in 2009, 62% had one complex chronic condition, and 43% had 3 or more chronic conditions, in addition to tracheostomy.[13] In the majority of these children, the tracheostomy may be required for a number of years, if not a lifetime, with an ongoing need for long-term, complex tertiary care and labor-intensive home care, especially in those children requiring chronic ventilatory support.

The decision to perform a tracheostomy is easy in patients requiring a temporary airway to bypass an anatomic or acquired airway obstruction or in those with chronic respiratory failure that will subsequently resolve. However, many patients have severe chronic comorbidities, including cerebral disabilities. Decision making in these patients may not be as straightforward.

Preoperative workup should always include a thorough physical examination to assess neck mobility preparing for the hyperextension necessary for the surgery. Coagulation and blood cell count studies should also be done previously. Further testing depends on each patient's comorbidities.

### *Tracheostomy interdisciplinary team*

The importance of an interdisciplinary team to approach complex children requiring a tracheostomy cannot be overstated. It should oversee the process from pre-operative discussions to long-term care and eventual decannulation. Patient safety and tolerance are central to these considerations. Aside from issues of safety, stoma care and ongoing management of airway pathology, many of these children have other physical, medical, developmental, communicational and psychological problems that must be addressed in parallel. In a series by Schweiger et al, up to 84% of the tracheostomized patients had important comorbidities.[14] This team should involve intensive care and pulmonology specialists, pediatric otolaryngologists, pediatric surgeons, social care workers, speech-language pathologists and respiratory nurses, taking into account the morbidities and even mortality associated with a long-term tracheostomy. The family should also be involved and counseled during this process. Abode et al demonstrated that a multidisciplinary approach can yield measurable improvements in important outcomes. Their study had 3 primary aims: optimize stay for children with newly placed tracheostomies, improve communication among providers and caregivers, and avoid tracheostomy complications. They succeeded in all of them, recommending that all institutions with pediatric airway programs should adopt this team approach for tracheostomized patients[15]

### *Choosing the right tracheostomy tube*

It is particularly important that the size of the tracheostomy tube that is selected is appropriate for both the size of the child's airway and the clinical indication for the procedure. Generally, the smallest tube that allows for adequate ventilation is chosen. An oversized tube may cause injuries in the tracheal walls, eventually leading to fistulization, and a tube that is too long may result in monobronchial cannulation. The appropriate size of the tracheostomy tube can be estimated by using the endotracheal tube formula for children proposed by Cole.[16] (internal diameter [ID] [mm] = [age in years/4] + 4.0) for uncuffed ETT and Motoyama (ID [mm] = [age in year/4] + 3.5) for cuffed ETT, although recent evidence points at the cuffed ETT formula being safer to avoid an oversized tube[17]. This can then be converted to the appropriately sized tracheostomy tube. (See Table 2)[18]

**Table 2**
ETT tube sizes.

| Age | Preterm | Term | 6m | 1 | 2 | 3 | 4 | 5 | 6 | 8 | 10 | 12 | 14 | Adult |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Avg weight (kg) | | 3.5 | 7 | 10 | 12 | 14 | 16 | 18 | 20 | 25 | 30 | 40 | 50 | 70 |
| ETT size (age + 16)/4 | 2.5-3 | 3-3.5 | 3.5-4 | 4 | 4.5 | 4.5 | 5 | 5 | 5.5 | 6 | 6.5 | 7 | 7 | 7.5-8 |

**Table 3**
Chart for Tracheostomy tube sizing.

| **All dimensions in mm** | | **Preterm-1 month** | | **1-6 months** | **6-18 months** | **18 months-3 years** | **3-6 years** | **6-9 years** | **9-12 years** | **12+ years** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Trachea (transverse diameter)** | | 5 | | 5-6 | 6-7 | 7-8 | 8-9 | 9-10 | 10-13 | 13 |
| **Shiley®** | **Size=ID** | | **3,0** | **3,5** | **4,0** | **4,5** | **5,0** | **5,5** | **6,0** | **6,5** |
| **Cuffless** | OD | | 4,5 | 5,2 | 5,9 | 6,5 | 7,1 | 7,7 | 8,3 | 9,0 |
| | Length Neo | | 30 | 32 | 34 | 36 | | | | |
| | Length Ped | | 39 | 40 | 41 | 42 | 44 | 46 | | |
| **Tracoe mini®** | **Size=ID** | **2,5** | **3,0** | **3,5** | **4,0** | **4,5** | **5,0** | **5,5** | **6,0** | |
| | OD | 3,6 | 4,3 | 5,0 | 5,6 | 6,3 | 7,0 | 7,6 | 8,4 | |
| | Length Neo | 30 | 32 | 34 | 36 | | | | | |
| | Length Ped | 32 | 36 | 40 | 44 | 48 | 50 | 55 | 62 | |
| **Tracoe Silcosoft®** | **Size=ID** | **2,5** | **3,0** | **3,5** | **4,0** | **4,5** | **5,0** | **5,5** | | |
| | OD | 4,4 | 4,7 | 5,4 | 6,0 | 6,6 | 7,3 | 8,4 | | |
| | Length Neo | 30 | 32 | 34 | 36 | | | | | |
| | Length Ped | 38 | 39 | 40 | 41 | 42 | 44 | 46 | | |

**Fig 1.** TRACOE silicone Silcosoft® cuffless and cuffed tubes on top. Bottom left, TRACOE mini PVC pediatric tube. Bottom right, Shiley Pediatric PVC tube.

There are many different manufacturers, models and sizes of tracheostomy tubes, each with their own advantages and disadvantages. In our institution, we use Tracoe and Shiley pediatric and neonatal tubes, which dimensions are summarized in Table 3. (Fig. 1)

There are two materials to choose from when deciding which tracheostomy tube to use: PVC (Polyvinyl chloride) or silicone. The main advantage of silicone over PVC is its unique capability of reducing the biofilm formation on its surface, thus reducing the presence of periostomal granulation tissue.

When choosing a cuffed tube, it is important to consider its diameter with the cuff deflated. If this diameter is bigger than the diameter of the shaft of the tube, insertion and tube change may become difficult. In this respect, TRACOE Silcosoft® cuffed tubes have an exceptionally low profile, which makes them much better than Shiley® cuffed tubes. They are also inflated with water, so changes in cuff diameter are not possible. Importantly, as the cuff is filled with water, high pressure is exerted on the tracheal wall. A manometer cannot be used with a water-filled cuff. To avoid damage, regular cuff deflations must be performed in order to protect the tracheal mucosa from pressure injury and subsequent stenosis risk. The frequency of cuff deflations varies according to inflation volumes, which must be clearly recorded, and cuff-use duration. Cuff deflation every 2 h for 5–10 min, if tolerated, is recommended. With regard to the amount of air or water a cuff should be inflated with, an air-filled cuff should be monitored with an appropriate manometer. For the water-filled cuff, the amount of water should be slowly titrated to a level that allows a minimal air leak.[19]

## Surgical technique

*Percutaneous tracheostomy*

Percutaneous tracheostomy has largely replaced surgical tracheostomy in adult patients. It is considered a safe and an easy bedside procedure that does not damage tracheal cartilages and, in addition, has better cosmetic results. However, this technique is rarely used in children, due to concerns about the safety of the procedure and technical limitations, especially in young children and infants. Almost 50% of the tracheostomies are performed in infants younger than 1 year of age, who have very small airways, making it very difficult, and in some cases impossible, to ventilate them through a small endotracheal tube occluded by a bronchoscope. There is also the concern for accidental decannulation in the early post-operative period, that can be fatal, due to the absence of stay sutures which are present in surgical tracheostomy to facilitate tube reinsertion. Finally, many of these patients are tracheostomized because they have an airway narrowing (congenital or acquired) that can make percutaneous tracheostomy very challenging, if not impossible.

Overall experience in children is very limited. Large series describing risks and benefits are lacking, longer follow-up and more patients are needed to determine the long-term benefits of the procedure in children, as well as to determine the lowest age for its safe performance. The choice of the author remains the traditional surgical tracheostomy due to this paucity of evidence.

*Surgical tracheostomy*

Before the procedure, we always perform a laryngotracheoscopy to assess the indemnity of the subglottis and the trachea, and to discard other eventual lesions which could need treatment after the tracheostomy is performed (glottic or subglottic granulation tissue, for example)

*Step 1. Patient placement*

The procedure is performed with the child lying supine on the operating table. Adequate exposure of the larynx is achieved by hyperextending the neck, which can be achieved by placing a jelly roll under the patient's shoulders. We prefer an inflated pressure infusion cuff, that allows to place the neck in a good position, and to be quickly deflated when the procedure is finished, or if it is needed during the procedure, diminishing the risk of field contamination that could take place when retiring a sandbag or jelly roll. (Fig. 2)

**Fig 2.** Patient positioning with hyperextended neck.

*Step 2. Identifying structures*

The neck is then palpated to identify the hyoid bone, the cricothyroid membrane, the inferior margin of the cricoid cartilage, and the suprasternal notch. These structures are marked on the skin as a guide. (Fig. 3)

**Fig. 3.** skin marks previous to tracheostomy..

*Step 3. Incision and defattening*

A horizontal neck incision is made midway between the suprasternal notch and the inferior margin of the cricoid cartilage. Subcutaneous fat is then removed with a monopolar cautery, taking care not to damage the anterior jugular veins, which can be medialized in some patients. This step is very useful to better expose deeper tissues, and to facilitate the replacement of an accidentally displaced or blocked tube in the immediate postoperative period.

*Step 4. Dissection*

Blunt dissection is then carefully performed until reaching the strap muscles, which are divided in the midline and retracted laterally, revealing the underlying thyroid isthmus and trachea. It is especially important to palpate the trachea frequently to avoid lateral dissection and injury to surrounding tissues. It is also important to cauterize any bleeding vessels to keep the surgical field clean.

*Step 5. Thyroid isthmus section*

The thyroid isthmus is separated from the trachea with curved forceps, and divided with a monopolar or bipolar cautery, thus exposing the cricoid and anterior tracheal wall. (Fig. 4)

*Step 6. Stay suture placement*

Prolene stay sutures are placed laterally to aid in the initial incision and are left in place until the first tracheostomy tube change. They also facilitate the recannulation of the trachea in the event of a displaced tube in the postoperative period. (Fig. 5)

*Step 7. Tracheal incision*

The tracheal incision is made vertically in the midline of the anterior tracheal wall, taking into account the future need for reconstructive surgery if the procedure is being done for a subglottic stenosis. It is better to place the tracheostomy very high (i.e. second and third rings) or very low (i.e. fifth and sixth rings) to avoid

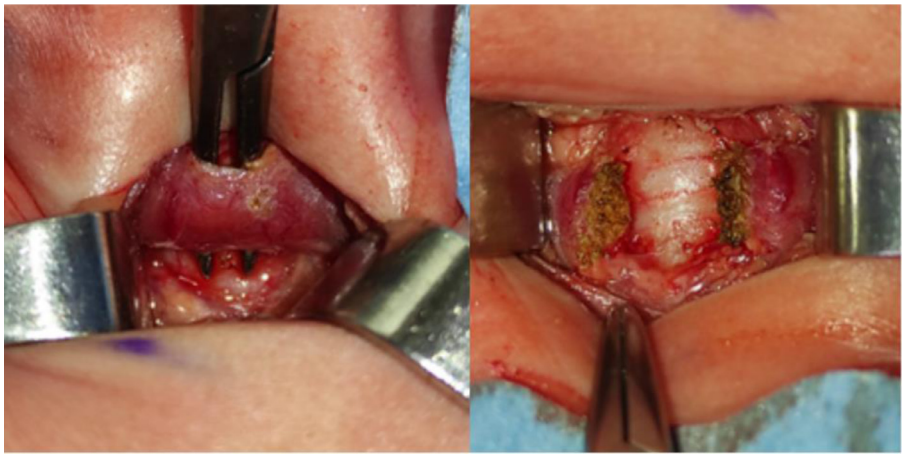

**Fig. 4.** Blunt dissection of the thyroid isthmus on the left. Isthmus incised in the midline on the right.

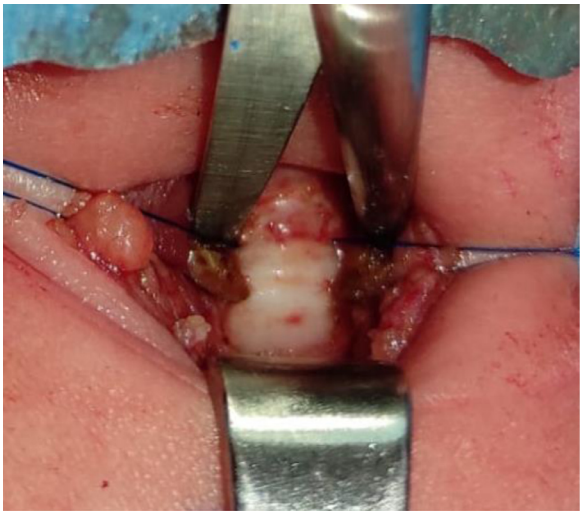

**Fig. 5.** Prolene stay sutures in place.

unnecessary resection of a healthy segment during the reconstructive surgery. If the procedure is being done for other reasons, it is better to incise the third and fourth rings to avoid future cricoid damage due to erosion by pressure from the tracheostomy tube. This is especially true for expected long-term tracheostomies in chronically ventilated or neurologically impaired patients. (Fig. 6)

Care should be taken at this point to avoid a fire in the airway and the operative field due to cautery usage and oxygen leaking from the open trachea. Communication with the anesthesiologist is fundamental at all times. Ideally, less than 30% oxygen should be delivered to the patient without nitrous oxide to minimize this risk, and an expired oxygen concentration should also be under 30%.[20,21]

An option to the vertical incision is the Björk flap, which is an inferiorly based flap that is sutured to the skin, and allows for easier reinsertion of the tracheostomy tube, should it be displaced in the immediate postoperative period. Figure[7]

*Step 8. Maturation sutures*

Maturation sutures are then placed around the stoma, tethering the borders to the skin using 4.0 or 5.0 resorbable sutures (the authors choice is PDS). It is important to keep these sutures as close to the borders of the skin (<1 mm) as possible. They provide an additional safety measure as they, again, allow for easier replacement of a displaced tracheostomy tube in the immediate postoperative period. (Fig. 8)

*Step 9. Tube insertion*

After the maturation sutures are in place, the anesthetist withdraws the endotracheal tube until the tip is just superior to the stoma. It is very useful to keep a sterile circuit and cuffless endotracheal tubes on the surgical tray. The latter are used to gently dilate the stoma to allow for an easy but snug insertion of the tracheostomy tube in case it is necessary. The sterile circuit is then connected to the tracheostomy tube and handed to the anesthetist, who in turn completely withdraws the endotracheal tube. (Fig. 9)

At this time, it is crucial to ensure that a capnography curve is present in the anesthesia monitor, and that there is symmetrical ventilation of both lungs after deflating the pressure infusion cuff, or any shoulder bolster used during the procedure, because the tip of the tracheostomy tube can migrate distally after relieving the neck hyperextension, resulting in monobronchial cannulation. A flexible tracheoscopy can also be made at this moment to ensure the position of the tip of the tracheostomy tube. At our institution, an immediate radiographic control is performed for this effect.

A non-adherent dressing is then applied to the skin around the tube, which is then fixed to the patient's neck with a velcro neck tie.

## Complications

The incidence of pediatric tracheostomy complications in the literature ranges from 0 to 90% according to different series. In a metanalysis by Lubianca et al.[22] they found an average close to 40% in 49 articles.

Complications have traditionally been classified according to the moment in which they present in intraoperative, early postoperative (during the first week), and late postoperative (after the first week). Table 4 summarizes the most frequent complications, some of which can also be seen in Fig. 10.

*Preventing complications*

*Intraoperative*

The cornerstone for the prevention of intraoperative complications is a careful surgical technique with meticulous hemostasis during the procedure. To avoid inadvertent damage to the esophagus, especially in prematures or very small newborns, it is advis-

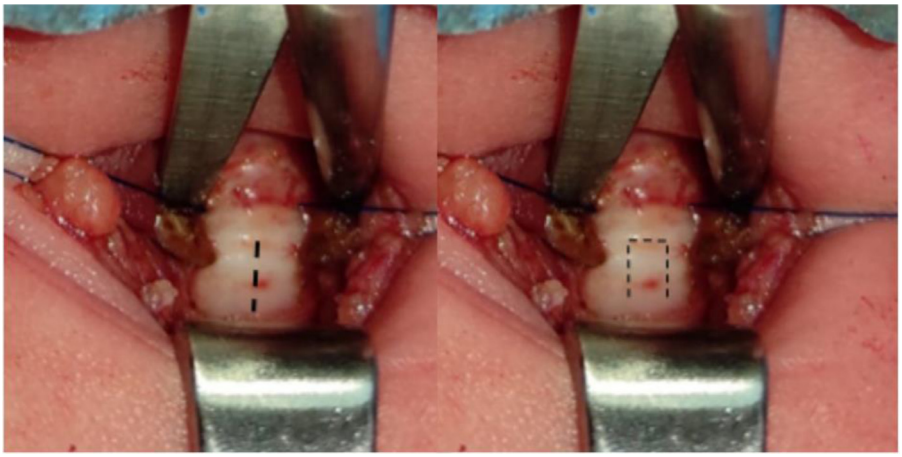

**Fig. 6.** Left: vertical tracheal incision. Right: Björk flap

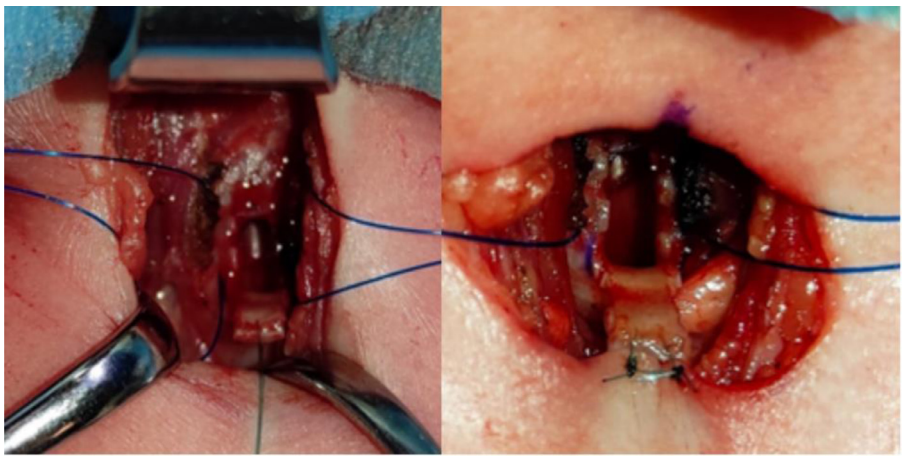

**Fig. 7.** Björk flap.

**Fig. 8.** PDS maturation sutures in place.

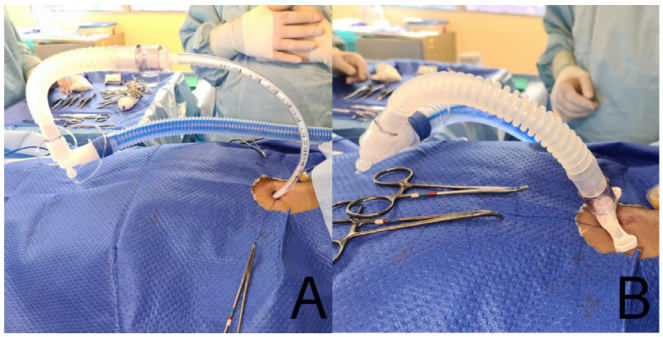


**Fig. 9.** A. Sterile circuit with an endotracheal tube used for gentle dilation of the stoma. B. Tracheostomy tube in place.

able to remove naso/orogastric tubes, or esophageal temperature probes prior to the surgery.

The choice of a correct sized tube is very important to avoid damaging the tracheal walls upon insertion. Airway fire precautions cannot be overemphasized. Communication must be maintained with the anesthetist at all times.

*Early postoperative*

Many of the early postoperative complications are also linked to an inadequate surgical technique. In this period, adequate nursing care also becomes very important. Skilled nursing care in an appropriate setting with 24-h surveillance is necessary. Correct tracheostomy tube care is key to preventing postoperative complications. In order to reduce the risk of the tracheostomy tube becoming clogged with dried secretions, frequent suctioning with instillation of saline solution and constant humidification of the inspired air are essential.[23]

**Table 4**
Tracheostomy complications.

| **Intraoperative complications** |
|---|
| Excess bleeding |
| Injury to surrounding structures (cricoid cartilage, esophagus, recurrent laryngeal nerve, brachiocephalic trunk, cervical pleura) |
| False passage |
| Posterior tracheal lacerations |
| Monobronchial intubation |
| Pneumothorax/pneumomediastinum |
| Airway fire |
| **Early postoperative complications** |
| Inadverted lung damage that can lead to subcutaneous emphysema, pneumothorax, pneumomediastinum |
| Hemorrhage |
| Accidental decannulation |
| Tube obstruction with blood or secretions |
| Stomal leak that prevents the correct ventilation of the patient |
| peristomal skin infection (cellulitis) |
| **Late postoperative complications** |
| Accidental decannulation |
| Mucus plugging |
| Local and lower airway infections |
| Structural cricoid or tracheal damage (Subglottic stenosis, suprastomal collapse, tracheal stenosis) |
| Tracheovascular fistula |
| Tracheoesophageal fistula |
| Peristomal granulomas |
| Stomal dehiscence |
| Distal tracheal granulomas |
| Tracheocutaneous fistula |
| Swallowing dysfunction |

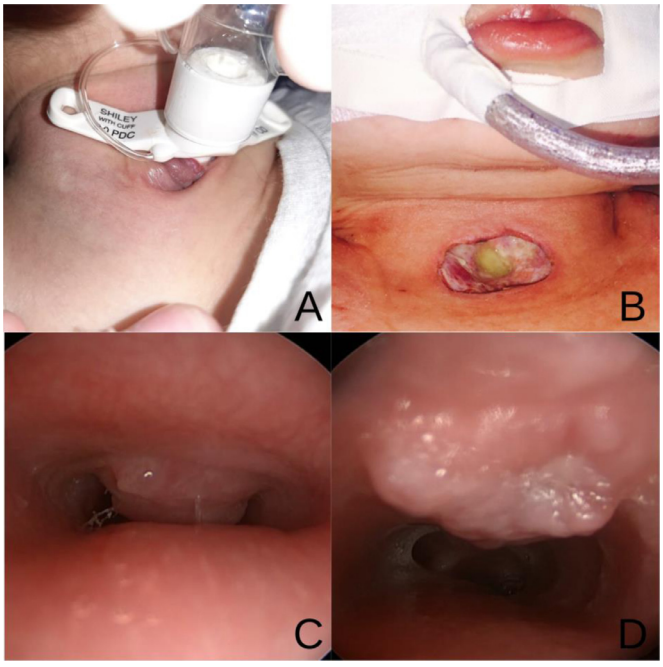


**Fig. 10.** Tracheostomy complications. A. Stomal granulomas. B. Stomal dehiscence and infection. C. Suprastomal collapse and granuloma. D. Distal tracheal granuloma at the tip of the tracheostomy tube.

*Late postoperative*

Pediatric tracheostomy, while increasingly considered a safe procedure for infants and children, requires significant long-term postoperative care, as most children who are eventually decannulated will still maintain their tracheostomy for around 2 years.[24]

Mortality typically reflects the underlying comorbidities necessitating tracheostomy, although tracheostomy-associated mortality is not inconsequential. In a literature review on tracheostomy-related complications, Dal'astra et al.[25] found that the overall mortality rate of pediatric tracheostomy ranged from 2.2% to 59%, with a tracheostomy-specific mortality of 0% to 5.9%.

Education of parents and caregivers as to the appropriate post-operative management is critical to reducing complications after tracheostomy. In a 2016 publication by Gaudreau et al,[26] they demonstrated that after applying a formal education protocol for caregivers, they achieved a significant reduction in tracheostomy-related wound complications (skin pressure ulcers, granulomas, infection).

**Table 5**
Adapted from IPOG recommendations for the management of pediatric tracheostomy in the early postoperative period.

| | |
|---|---|
| **Day 0-4** | |
| Airway | Skin care |
| Wean ventilator to goal of tracheostomy collar or baseline settings if chronically ventilated, unless the patient is chemically paralyzed | Assess neck skin Q4H and PRN for signs and symptoms of infection or breakdown |
| Humidification | Contact ORL if dressing needs to be changed or ties are too tight or too loose |
| Measure appropriate suction length | Head position: midline/sniffing position |
| Suction Q4H and PRN | When moving patient, take precautions to keep head and body aligned |
| Spare trach (same size and one size smaller) at bedside | Secure/support tracheostomy tube; prevent tension/pulling from trach collar or ventilator tubing |
| Information sheet at bedside | |
| **Day 5-7** | |
| Airway | Skin care |
| First tracheostomy tube change | Change tracheostomy ties daily or PRN when wet |
| Headlight, shoulder roll, flexible suction, and consider flexible tracheoscopy | Check under tracheostomy ties for signs of breakdown, irritation or rash |
| Assess need for cuffed, flextend or custom tracheostomy tube | Clean stoma daily/PRN |
| Tracheostomy tube profile sheet with updated tracheostomy/suction information | Apply tracheostomy dressing as needed |
| Spare trach (same size and one size smaller) along with information sheet at bedside | Check tie tightness while patient is sitting or side-lying every shift; only one small finger should fit between neck and tie |
| **Post-tracheostomy tube change** | |
| Airway | Skin care |
| Suction Q4H/PRN | As above for days 5-7 |
| Tracheostomy tube teaching for caregivers | |
| Spare trach (same size and one size smaller) along with information sheet at bedside | |

## Post operative care

*Early postoperative care*

In 2016, the International Pediatric Otolaryngology Group issued recommendations for the post-operative care of tracheostomized patients in the early postoperative period.[27] They made their recommendations based on the experts opinion on 4 main considerations: airway, sedation, skin care and nutrition. Sedation and nutrition should be discussed with the ICU team, and it is beyond the scope of this article to discuss the recommendations in depth.

Table 5 summarizes the experts opinion on airway and skin care. They also recommend considering an intraoperative flexible tracheoscopy to ensure the position of the tracheostomy tube tip. In our institution, we use immediate radiologic control for this purpose.

Antibiotics should be given in all patients until the first tracheostomy tube change.

**Table 6**
Benefits of early first tracheostomy tube change

| |
|---|
| Shorter length of ICU stay |
| Reduced cost |
| Lower requirement for heavy sedation |
| Less complications from withdrawal of sedation |
| Allows earlier wound assessment and reduces potential wound complications |
| Earlier placement of fresh tracheostomy ties |
| Earlier start to parent/caregiver training |
| Earlier patient mobility and rehabilitation |

**Table 7**
Home care educational components

| |
|---|
| Teaching component |
| Signs of respiratory distress |
| Signs of infection and skin breakdown |
| Checklist of emergency supplies |
| Contacts |
| Health care provider |
| Pertinent health care personnel |
| Equipment supply company |
| Hands-on component |
| When and how to suction the tracheostomy tube |
| When and how to clean the tube and the surrounding area |
| When and how to change tracheostomy tube |
| Use of home equipment |

It is also recommended, as a safety measure, to always have a tracheostomy sheet at the bedside with information of the patient, the tracheostomy tube (brand, type, dimensions), suctioning distance, patency of the airway above the tracheostomy, and the dates of the procedure and tube changes.

There is not wide consensus about the timing of the first tracheostomy tube change. Most authors consider days 5–7 to be adequate. In 2019, however, Woods et al[28] demonstrated that an early tube change (2-3 days) may be safe, and can have many benefits, which are shown in Table 6. In our institution, the first tube change is done by a member of the airway team on postoperative day 7.

Regardless of the time it is made, the first tracheostomy tube change has to be done in an ICU setting, with a headlight, a shoulder bolster, flexible suction, and the same tube used during the surgery, unless a special tube is required due to the ventilatory condition of the patient. At this time, the non-resorbable stay sutures are removed.

### *Late postoperative care*

There is great variation among different centers in the practice of home tracheostomy care, as was shown by a recent survey published by the International Pediatric Otolaryngology Group.[29] They found significant differences in diverse topics such as inpatient teaching of the parents, home care, tracheitis evaluation and management and tracheostomy surveillance.

There is consensus, however, in the need of proper education for the parents/caregivers as a fundamental element to lessen long-term complications such as tube obstruction and wound and tracheal infection. Table 7 shows the components that should be taught to all caregivers for tracheostomy care at home, as recommended by a 2013 clinical consensus statement by the American Academy of Otolaryngology-Head and Neck Surgery[30]

One safety aspect is ensuring that families have access to the necessary emergency equipment. In 2019, Kohn et al. published their work detailing the use of “Go-bags” to standardize delivery of emergency supplies to children with tracheostomies[31]. A “Go-bag” is a standardized compact bag that contains all the necessary supplies to care for a patient with a tracheostomy. Included in the bag are spare tracheostomy tubes, obturator, suction supplies, saline, ties, dressing supplies, syringe, lubricant, gloves, scissors, and hand sanitizer. They found fewer tracheostomy related adverse events in the patients with “Go-bags” than in those without (22% vs 39% respectively), although with no statistical significance.

### *Surveillance endoscopy*

There is no standardized protocol for airway surveillance in pediatric tracheostomy patients. Some institutions perform routine endoscopic surveillance of patients following tracheostomy, whereas others only perform endoscopic evaluations when the patients exhibit symptoms of airway pathology (bleeding, difficult tracheostomy tube changes, ventilator dependence, speaking valve intolerance) A survey of pediatric otolaryngologists found that most practitioners perform at least yearly surveillance endoscopy in children under age 2 years, but many only perform endoscopy on patients prior to decannulation or in those experiencing difficulties.[32] A recent series by Liu et al[33] showed a high rate (83% to 91% in the first three airway endoscopies) of abnormal airway findings in a cohort of 127 tracheostomized patients. Their most common findings were subglottic stenosis, suprastomal granulation tissue, glottic edema, and suprastomal collapse. Up to 33% of these patients eventually require surgical treatment for the endoscopic findings,[34] which supports the need for routine airway surveillance even in the patients with no difficulties. The author performs a laryngotracheoscopy every six months in all tracheostomized patients.

## Decannulation

Decannulation as soon as the child’s underlying conditions permit is the ultimate goal shared by patient, family and physician alike. It reduces tracheostomy related complications, negative impacts on patient and family quality of life, and the socioeconomic burden of a tracheostomy on patients and the health care system.

The literature shows a rate of decannulation failure ranging from 8 to 22,3%. To minimize this risk, there are many protocols that use varying combinations of in-patient/out-patient resources, specialized tests, and procedures. This has led to substantial variability in the duration of hospitalization for patients undergoing decannulation.

The International Pediatric Otolaryngology Group has recently issued a paper on recommendations for pediatric tracheostomy decannulation.[35] They discuss some issues that should be considered when attempting to decannulate a patient, such as readiness to decannulate, steps to decannulation, decannulation protocol, and decannulation failures.

### *Readiness to decannulate*

According to the recommendations by the International Pediatric Otolaryngology Group, there are 7 major factors to consider when assessing a child’s readiness for decannulation. These are (in order of importance): Airway endoscopy findings, oxygenation status, secretion management, comorbidities, level of consciousness, respiratory rate and sleep study findings. In the 2013 consensus[30], the authors also recommend that the child has not been under ventilatory support for at least 3 months. The patient should always have an awake transnasal fiberoptic laryngoscopy to assess vocal fold mobility and a complete dynamic and rigid airway examination to evaluate the airway anatomy and eventual dynamic collapse due to localized tracheomalacia at the stoma site. Any findings in the airway endoscopy that may preclude decannulation should be dealt with promptly before moving on to the next phase.

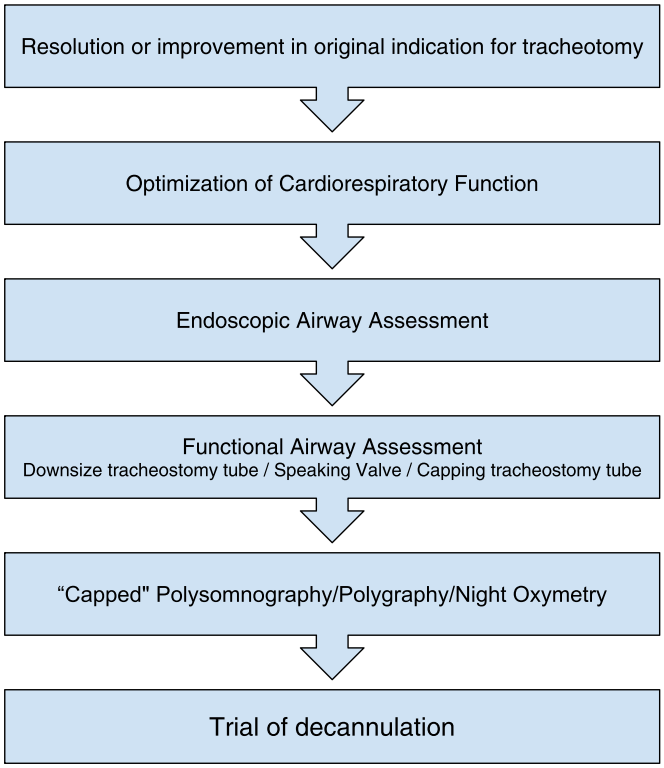


**Fig. 11.** Decannulation algorithm. Modified from Lee et al.

*Steps to decannulation and decannulation protocol*

Again, there is great variability in the literature on the steps that should be taken to lead to a successful decannulation. Nevertheless, all authors agree that a progressive approach should be made to minimize the risk of decannulation failure. Lee et al.[36] propose a simple algorithm (see Fig. 11) that is very similar to the protocol the author uses. Decannulation should be done in a hospitalary setting during an inpatient stay, in which a capped sleep study is recommended. In our institution we do not use a polysomnography in every patient, but rather do a polygraphy, or a night O2 saturometry with the tube capped.

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